COFILTERED NEIGHBOURHOODS MAKE THE STALK FUNCTOR LEFT EXACT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every step the author writes out is inside the counted span and no closing sentence is a placeholder.

Editorial note. Every labelled reference occurring inside the counted spans is carried in assembly.external_references as a non-load-bearing navigation pointer (same-chapter ones at their definition site, cross-chapter ones at the invocation inside the counted span), so no load-bearing mathematics is replaced by a stub. No bibliography citation occurs inside the counted spans.

Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Limits of sheaves are computed as presheaves. Same chapter, Lemma lemma-limit-sheaf. Identifies the limit of a finite diagram of sheaves with the presheaf limit.
- The description of limits of presheaves. Same chapter, the section describing limits and colimits of presheaves. Gives the explicit description of the presheaf limit used in the computation.
- Filtered colimits commute with finite limits. Categories, Lemma categories-lemma-directed-commutes. Supplies the commutation used to interchange the limit and the colimit defining the stalk.

Source reference. Original equation 04EH: sources/stacks-sites/pinned-original-source.tex, source lines 8458–8458. Navigation pointer invoked at proof line 8458; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 00W2: sources/stacks-sites/pinned-original-source.tex, source lines 8468–8468. Navigation pointer invoked at proof line 8468; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:sites.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original section 00VB: sources/stacks-sites/pinned-original-source.tex, source lines 8473–8473. Navigation pointer invoked at proof line 8473; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 002W: sources/stacks-sites/pinned-original-source.tex, source lines 8479–8479. Navigation pointer invoked at proof line 8479; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $p = u : \mathcal{C} \to Sets$ be a functor. If the category of neighbourhoods of p is cofiltered, then the stalk functor (04EH) is left exact.*

Proof. Let $\mathcal{I} \to Sh(\mathcal{C})$, $i \mapsto \mathcal{F}_i$ be a finite diagram of sheaves. We have to show that the stalk of the limit of this system agrees with the limit of the stalks. Let $\mathcal{F}$ be the limit of the system as a *presheaf*. According to Lemma 00W2 this is a sheaf and it is the limit in the category of sheaves. Hence we have to show that $\mathcal{F}_p = \lim_{\mathcal{I}} \mathcal{F}_{i,p}$. Recall also that $\mathcal{F}$ has a simple description, see Section 00VB. Thus we have to show that

$$\lim_i \operatorname{colim}_{\{(U,x)\}^{opp}} \mathcal{F}_i(U) = \operatorname{colim}_{\{(U,x)\}^{opp}} \lim_i \mathcal{F}_i(U).$$

This holds, by Categories, Lemma 002W, because the opposite of the category of neighbourhoods is filtered by assumption. □